

Pizza Sales Performance and Operational Analysis Report

Prepared by Tien Vuong Nguyen
Power BI Dashboard Analysis

1. Executive Summary

This report analyses pizza sales performance, customer purchasing behaviour, product-level demand, time-based order patterns, and operational turnaround indicators. The analysis is based on the completed Power BI dashboard pages and focuses on translating visual results into practical business insights.

Total Revenue	Total Orders	Total Quantity Sold	Average Order Value	Avg Pizzas / Order
\$817.86K	21,350	49,574	\$38.31	2.32

The business generated \$817.86K in total revenue from 21,350 orders and 49,574 pizzas sold. Average order value was \$38.31, and customers purchased an average of 2.32 pizzas per order. This indicates that revenue was supported not only by order frequency, but also by customers commonly buying multiple pizzas in one transaction.

1.1 Revenue by Pizza Category

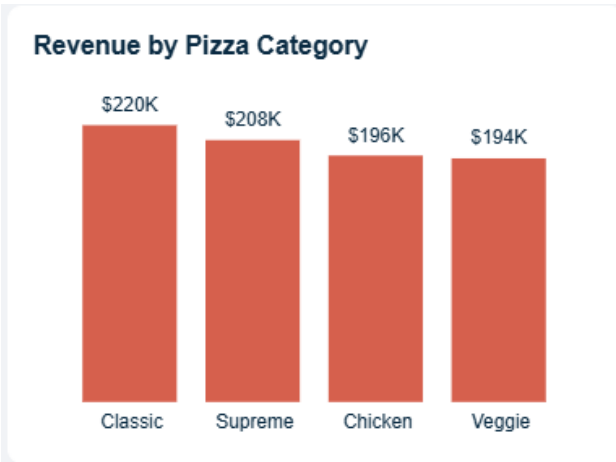


Fig 1. Revenue by pizza category

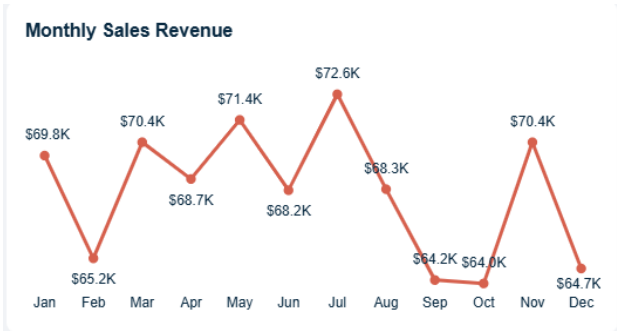


Fig 2. Monthly sales revenue

Classic generated the highest category revenue at \$220K, followed by Supreme (\$208K), Chicken (\$196K), and Veggie (\$194K). The difference between the highest and lowest categories was only around \$26K, showing that revenue was relatively diversified across the menu rather than dependent on a single category.

Monthly revenue fluctuated across the year. The highest revenue was recorded in July (\$72.6K), followed by May (\$71.4K), March (\$70.4K), and November (\$70.4K). The lowest months were October (\$64.0K), September (\$64.2K), and December (\$64.7K), suggesting that revenue was affected by short-term seasonal or behavioural factors rather than a consistent upward or downward trend.

1.2 Top Revenue Products

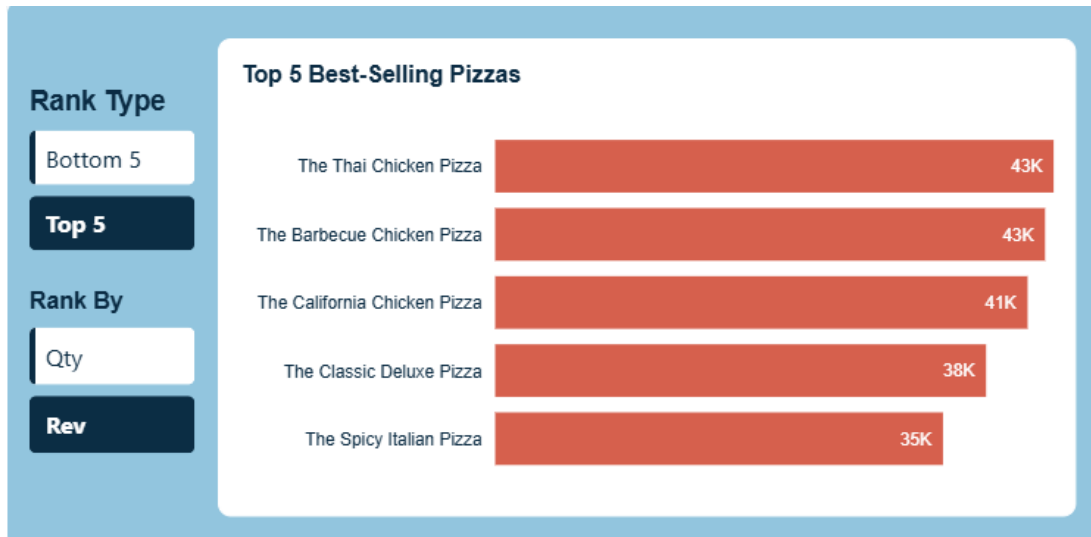


Fig 3. Top 5 best-selling pizzas by revenue

The Thai Chicken Pizza and The Barbecue Chicken Pizza were the highest revenue-generating products, each contributing approximately \$43K. The California Chicken Pizza followed with about \$41K. This confirms that Chicken pizzas played a major role in top product revenue, even though Classic was the strongest category by total quantity sold.

Key takeaway: overall performance was stable and diversified. Classic pizzas drove the largest category revenue and volume, while Chicken-based products dominated the top revenue ranking.

2. Monthly Sales Analysis: March Case Study

March is used as a representative monthly case study because it was one of the stronger months in the year, generating \$70.40K in revenue. The month also showed higher performance than the previous month across revenue, orders, and quantity sold.

March Revenue \$70.40K	March Orders 1,840	March Quantity Sold 4,261
---------------------------	-----------------------	------------------------------

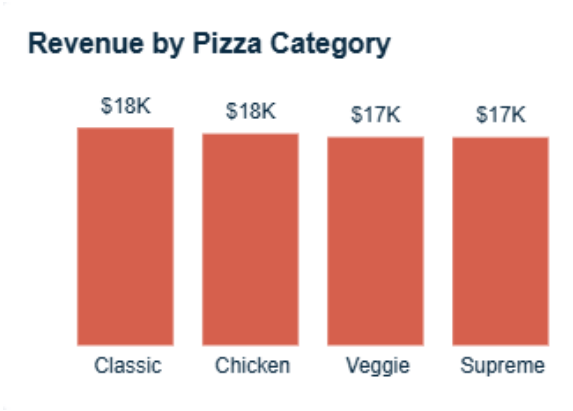


Fig 4. March revenue by category

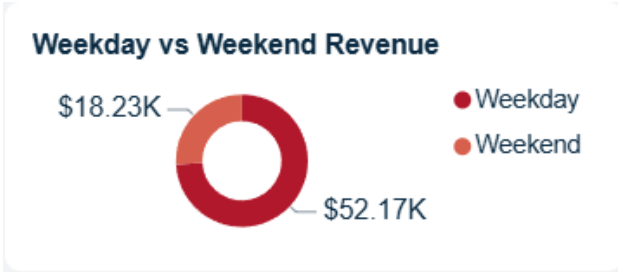


Fig 5. March weekday vs weekend revenue

In March, Classic and Chicken each generated approximately \$18K in revenue, while Veggie and Supreme each generated around \$17K. The small gap between categories indicates a balanced monthly revenue mix, with no single category fully dominating the month.

Weekend revenue contributed \$52.17K, compared with \$18.23K from weekdays. This confirms that weekend trading was the main driver of March revenue and should be a key focus for staffing, preparation planning, and ingredient stock availability.

2.1 March Product and Daily Revenue Patterns

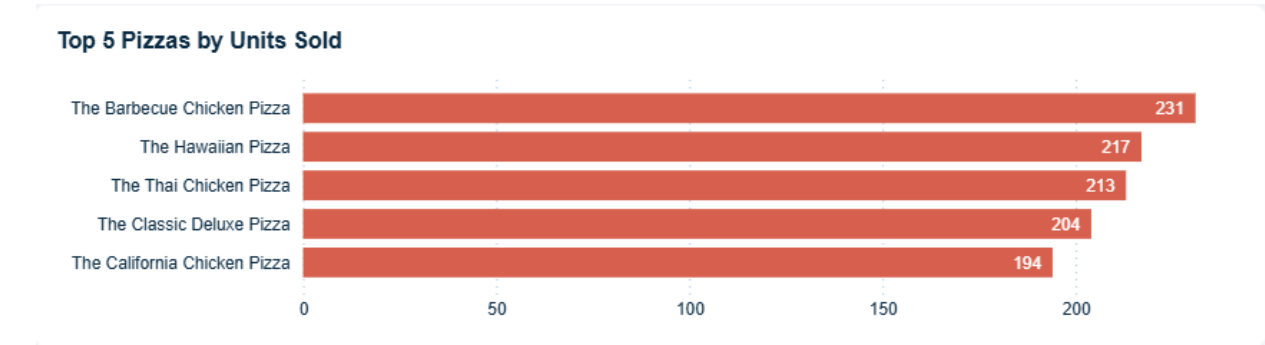


Fig 6. Top 5 pizzas by units sold in March

The Barbecue Chicken Pizza was the best-selling product in March, with 231 units sold. It was followed by The Hawaiian Pizza (217), The Thai Chicken Pizza (213), The Classic Deluxe Pizza (204), and The California Chicken Pizza (194). The narrow difference between the top five products shows that demand was spread across multiple popular pizzas rather than concentrated in one product.

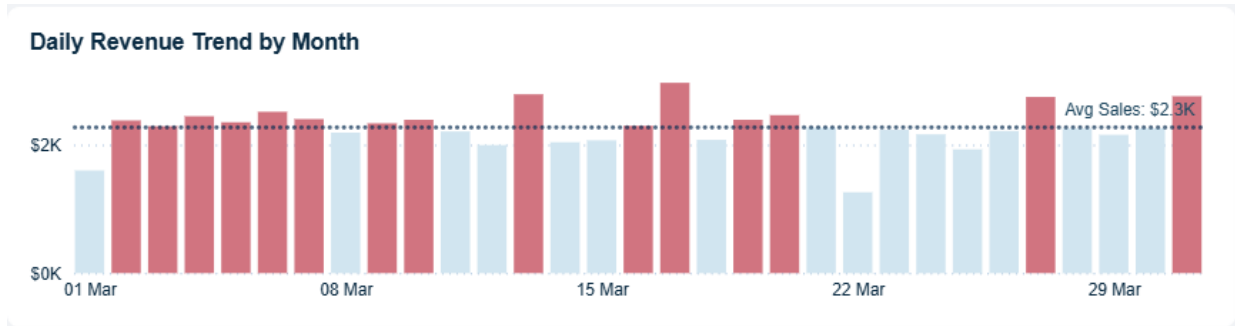


Fig 7. Daily revenue trend in March

Daily revenue in March was relatively stable around the average sales level of approximately \$2.3K per day. Several days performed above average, but the overall pattern remained consistent, indicating predictable customer demand during the month.

3. Sales Trends and Anomalies

3.1 Daily Orders and Moving Average

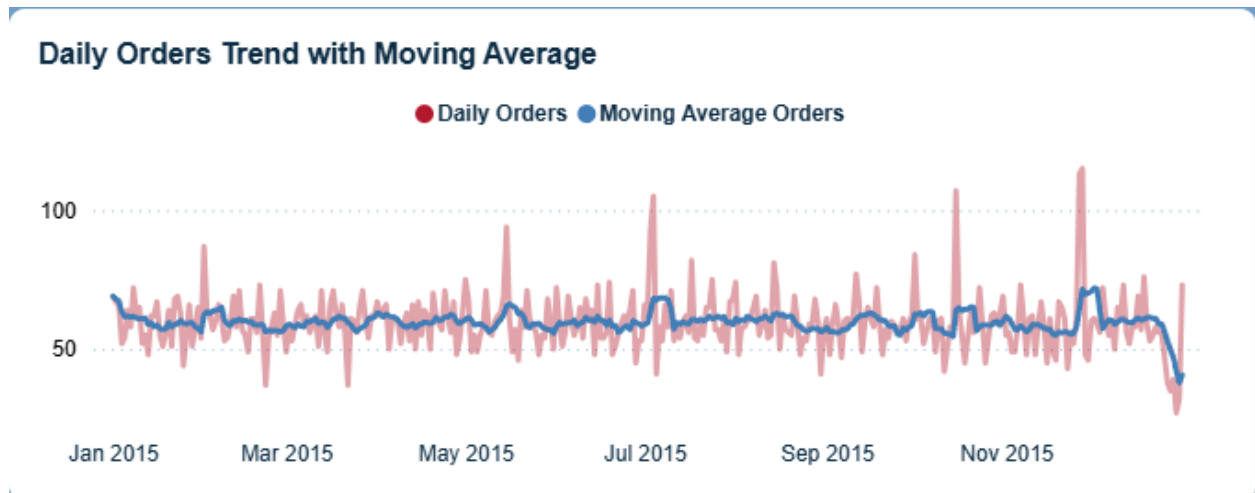


Fig 8. Daily orders trend with 7-day moving average

Daily order volumes fluctuated throughout 2015, but the 7-day moving average remained relatively stable around a baseline of approximately 60 orders per day. This indicates that the business had consistent underlying demand even though daily results varied.

The highest daily spike occurred on 27 November 2015, when orders reached 115. In contrast, the sharpest drop occurred on 29 December 2015, when orders fell to 27. These anomaly dates are important because they may reflect special events, promotions, holiday effects, or temporary operational changes.

3.2 Monthly and Weekly Order Behaviour

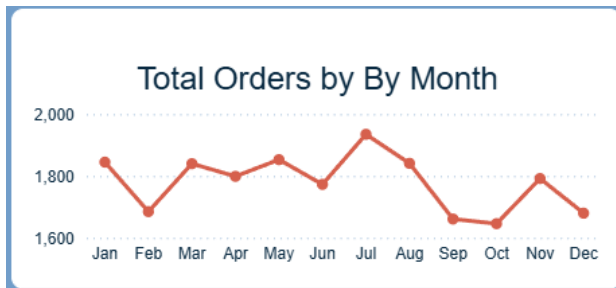


Fig 9. Total orders by month

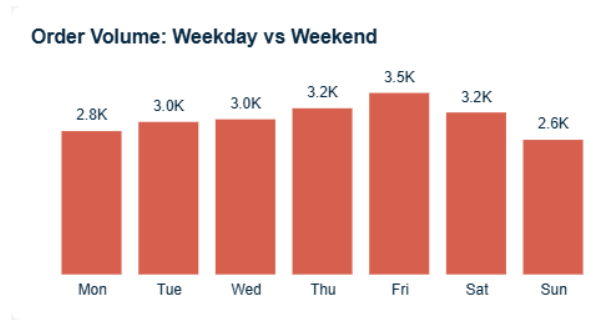


Fig 10. Order volume by weekday

Monthly order volume peaked in July at just under 2,000 orders. Orders declined from January to February, recovered in March, and remained relatively strong until August. September and October were the weakest months, before demand improved again in November.

Order volume was highest on Friday at approximately 3.5K orders, followed by Thursday and Saturday at around 3.2K each. Sunday recorded the lowest order volume at approximately 2.6K. This pattern confirms that demand increases toward the end of the week and is strongest around weekend trading periods.

3.3 Order Activity by Day and Hour

Order Activity by Day and Hour

	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Sun		1	95	210	261	214	222	278	322	324	281	197	147	70	2
Sat		1	85	222	260	225	235	294	325	388	371	328	265	147	12
Fri			192	415	413	221	190	258	344	387	340	319	268	180	11
Thu		3	214	434	438	233	228	289	376	361	270	217	114	60	2
Wed		2	231	420	381	201	210	259	343	335	238	193	139	72	
Tue	1		188	421	371	196	191	295	309	301	274	214	131	81	
Mon		1	226	398	331	182	192	247	317	303	235	174	134	53	1

Fig 11. Order activity by day and hour

The heatmap shows that peak order activity occurred between 12:00 PM and 8:00 PM, aligning with lunch and dinner periods. The strongest demand was concentrated in the evening, particularly around 6 PM to 8 PM. Friday and Saturday showed the highest activity, while Sunday and early weekdays were comparatively lower.

Key takeaway: although overall demand was stable, the business experienced clear daily, weekly, and hourly peaks. These patterns should guide staffing levels, kitchen preparation, and ingredient replenishment schedules.

4. Product, Category, Size and Ingredient Insights

4.1 Quantity Sold by Category

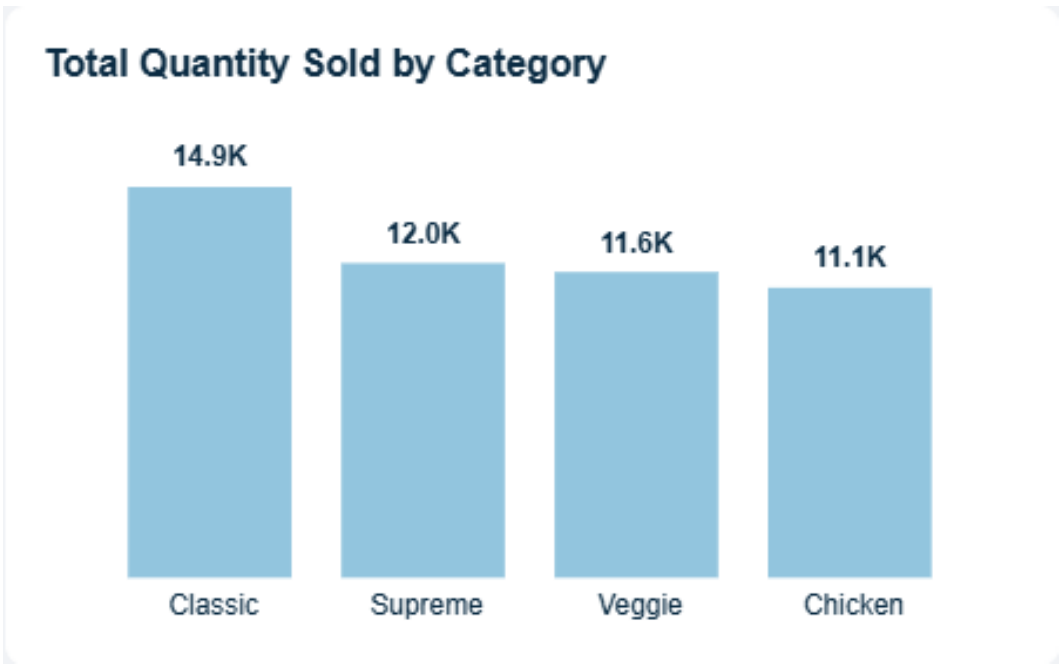


Fig 12. Total quantity sold by category

Classic had the highest sales volume, with approximately 14.9K pizzas sold. Supreme ranked second at 12.0K, followed by Veggie at 11.6K and Chicken at 11.1K. Classic therefore sold around 3.8K more pizzas than Chicken, confirming that it was the strongest category by unit volume.

4.2 Pizza Size Sales Distribution

Pizza Size Sales Distribution

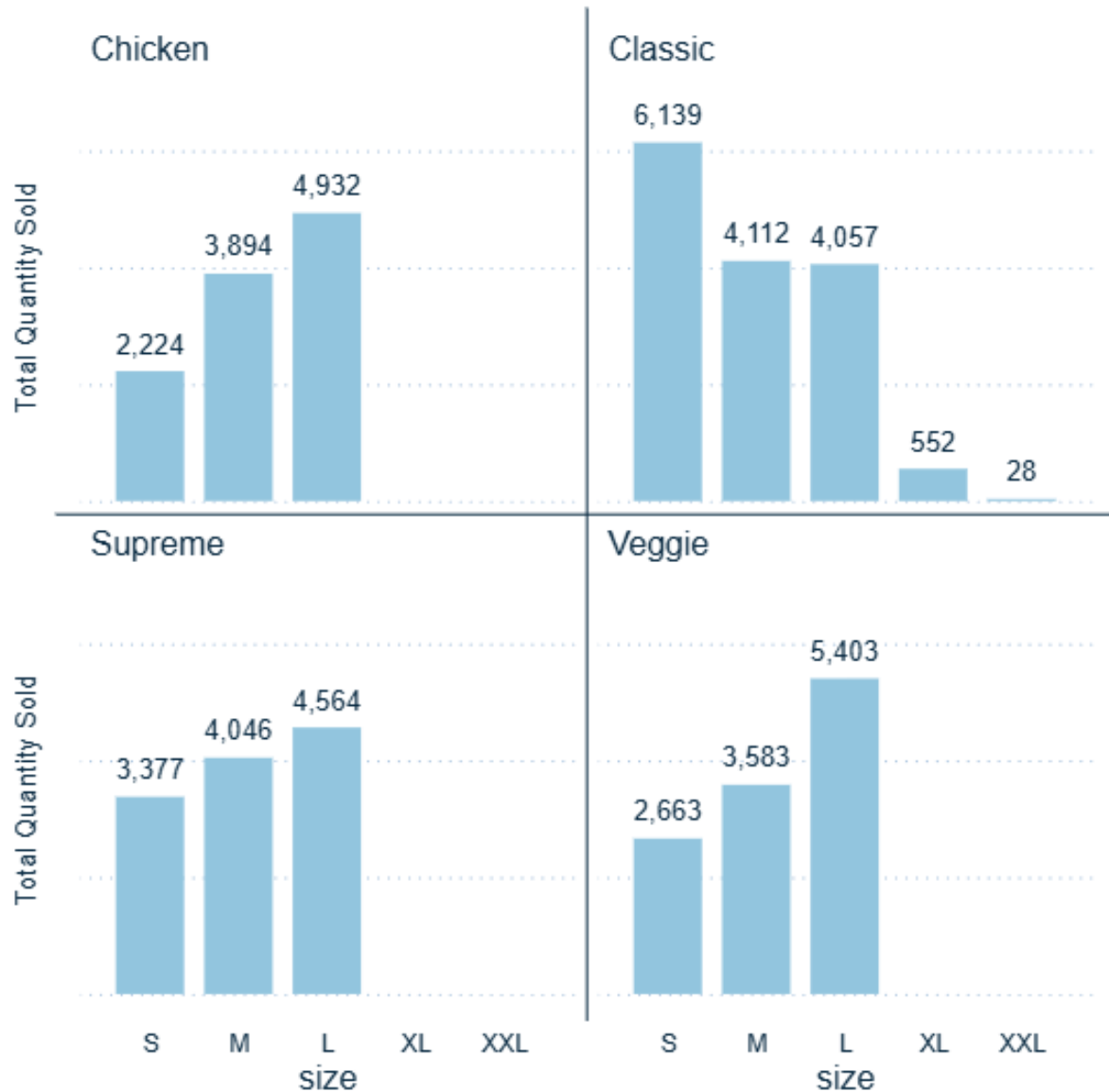


Fig 13. Pizza size sales distribution by category

Large pizzas were the most popular size in most categories. Chicken, Supreme, and Veggie all recorded their highest sales in Large size, with Veggie Large leading at 5,403 units. Classic followed a different pattern, where Small was the strongest size with 6,139 units, ahead of Medium and Large. Classic was also the only category with XL and XXL sales.

4.3 Ingredient Utilisation: Chicken Category

Top 5 Ingredients in Chicken Category

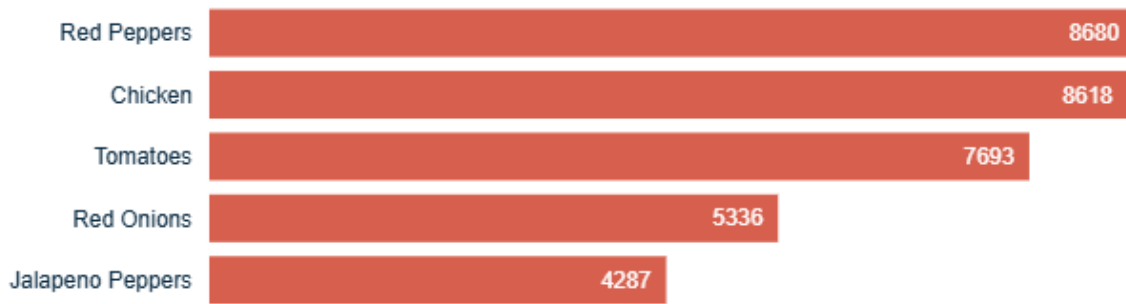


Fig 14. Top 5 ingredients in the Chicken category

Within the Chicken category, Red Peppers was the most used ingredient, associated with 8,680 units sold. Chicken followed closely at 8,618 units, and Tomatoes ranked third at 7,693 units. Red Onions and Jalapeno Peppers also contributed strongly. This suggests that top-performing Chicken pizzas combine protein-based toppings with strong vegetable ingredients.

Key takeaway: product demand is category-specific. Classic leads in quantity sold, while ingredient-level analysis helps identify which toppings support the strongest category performance.

5. Operational Performance Analysis

5.1 Turnaround Time Overview

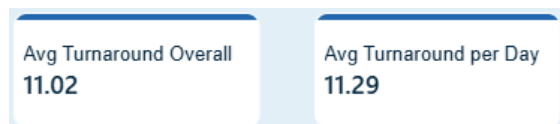


Fig 15. Average turnaround KPI cards

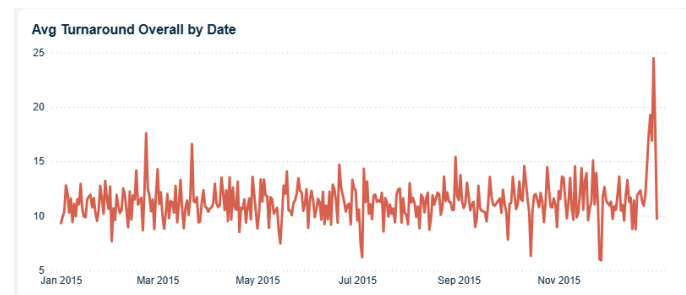


Fig 16. Average turnaround over time

Operational turnaround time remained relatively stable, with an overall average of 11.02 minutes and a daily average of 11.29 minutes. Most days were clustered around the 10-12 minute range, although several spikes appeared across the year. These spikes may be linked to demand surges, staffing constraints, or preparation bottlenecks.

5.2 Order Structure and Turnaround Breakdown

Single-Item vs Multi-Item Order Distribution

● Single-item Orders ● Multi-item Orders



Fig 17. Single-item vs multi-item order distribution

Order Turnaround Time Breakdown

● Zero Turnaround Orders ● NonZero Turnaround Orders

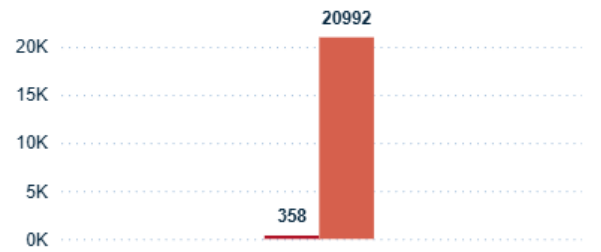


Fig 18. Zero vs non-zero turnaround orders

Multi-item orders accounted for approximately 13K out of 21K total orders, while single-item orders accounted for around 8K. This supports the earlier finding that customers commonly place larger orders, which increases average order value but also adds kitchen coordination complexity.

Most orders recorded non-zero turnaround time, with 20,992 non-zero turnaround orders compared with only 358 zero-turnaround orders. The low share of zero-turnaround records suggests that the dataset is operationally reliable for analysing preparation and fulfilment patterns.

5.3 Price vs Volume Relationship

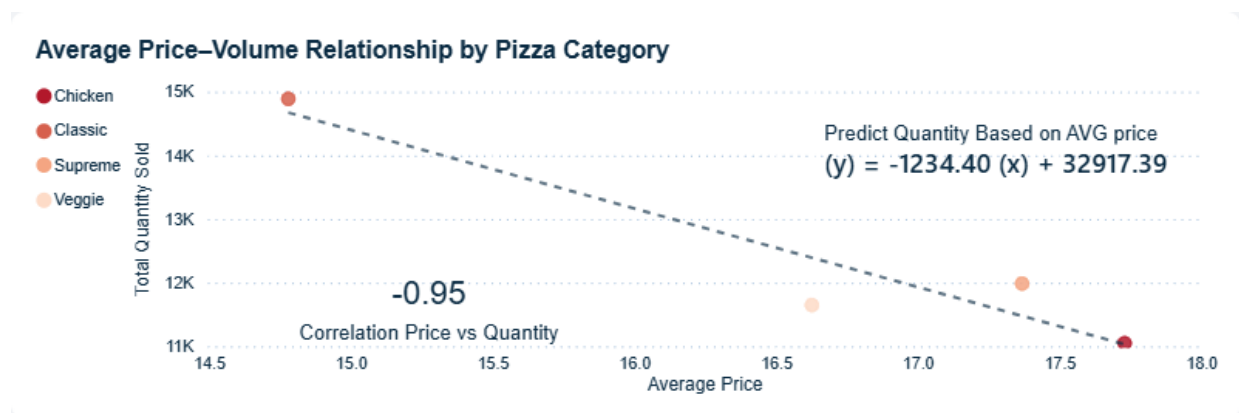


Fig 19. Average price-volume relationship by pizza category

The price-volume chart shows a strong negative relationship between average pizza price and quantity sold across categories. The correlation coefficient is approximately -0.95, meaning higher-priced categories tended to have lower sales volume. Classic had the lowest average price and the highest quantity sold, while Chicken had the highest average price and the lowest quantity sold.

The regression equation is $y = -1234.40x + 32917.39$. This suggests that each \$1 increase in average price is associated with an estimated decrease of about 1,234 pizzas sold. However, this result should be interpreted carefully because it is based on only four category-level data points.

6. Final Business Recommendations

- Prioritise weekend staffing and preparation capacity, especially because weekend revenue significantly exceeds weekday revenue in the March case study.
- Maintain strong availability of Classic pizzas because they are the leading category by both revenue and quantity sold.
- Use Chicken pizzas as premium revenue drivers, since Chicken products dominate the top revenue ranking.

- Plan inventory around peak ingredients such as Red Peppers, Chicken, Tomatoes, Garlic, and category-specific leading toppings.
- Monitor anomaly dates and daily spikes to identify potential promotional effects, holiday impacts, or operational bottlenecks.
- Use turnaround-time spikes to investigate kitchen workload, staffing coverage, and batching efficiency during peak order periods.

Overall, the dashboard indicates a stable pizza business with diversified revenue, clear weekend demand, strong Classic category volume, premium revenue contribution from Chicken products, and a generally predictable operational turnaround profile.